

Notes: de Bresser (RES, 2024)

Evaluating the Accuracy of Counterfactuals: Heterogeneous Survival Expectations in a Life Cycle Model

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Data source and sample	2
2.2	Institutional setting and pension reform	3
2.3	Subjective survival expectations	3
2.4	Three survival belief specifications	3
2.5	Life-cycle model outcomes	3
3	Models and empirical specifications	4
3.1	Retirement choices	4
3.2	Utility from consumption and leisure	4
3.3	Leisure, work costs, and stigma costs	4
3.4	Bequest utility	5
3.5	Health transition	5
3.6	Budget constraint	5
3.7	Measurement model for subjective survival	5
3.8	Estimation and validation	6
4	Identification and empirical design	6
4.1	Why expectations matter for identification	6
4.2	External validation through pension reform	6
4.3	What makes the heterogeneous-expectations model different	6
4.4	Limitations	7
5	Tables and figures	8
5.1	Figure 1 and Table 1: decisions and survival data	8
5.2	Figure 2: measurement model for subjective survival	8
5.3	Table 2: preference estimates	8
5.4	Figure 3: within-sample model fit	9
5.5	Table 3 and Figure 4: counterfactual pension reform predictions	10
5.6	Figure 5: behaviour conditional on survival beliefs before the reform	11
5.7	Figure 6: behaviour conditional on survival beliefs after the reform	11
5.8	Figure 7: why preference estimates matter for predictions	12
6	Short summary	12
7	Conclusion	13
7.1	Core conclusion	13
7.2	Economic intuition	13
7.3	Takeaway	13

1 Big picture

- **Question.** Do individual-specific subjective survival expectations improve the out-of-sample counterfactual predictions of a structural life-cycle model of retirement and saving?
- **Core identification issue.** Preferences and beliefs are jointly reflected in observed choices. If beliefs are misspecified, preference estimates can adjust to fit the same pre-reform data but generate wrong counterfactual predictions.
- **Setting.** The paper studies Dutch men aged 50 and older. The model is estimated on a pre-reform period, 1993–2001, when occupational pensions could be claimed from age 59 without actuarial penalties.
- **Policy validation.** The model is tested out of sample using the post-reform period, 2006–2016, when occupational pensions became actuarially adjusted and early retirement became much less generous.
- **Three beliefs specifications.** The paper compares life-table survival, average subjective survival, and heterogeneous subjective survival estimated from reported probabilities.
- **Main within-sample result.** All three models fit pre-reform labour supply, benefit claiming, and wealth profiles similarly well.
- **Main preference result.** Heterogeneous expectations change estimated preferences: risk aversion is higher, the discount factor is lower and more plausible, and the consumption weight is lower than in models with homogeneous expectations.
- **Main counterfactual result.** Models with homogeneous expectations predict too large a delay in retirement after the pension reform: about 4–5 years. The observed increase is 2.6 years. The heterogeneous-expectations model predicts 2.7 years.
- **Mechanism.** Heterogeneous survival expectations change how preferences are estimated and how workers with different beliefs react to actuarial adjustments.
- **Economic takeaway.** Structural models can fit observed behaviour under one regime while giving wrong policy predictions if expectations are misspecified. Subjective expectations are useful not just for estimation but for external validity.

2 Data and measurement

2.1 Data source and sample

- The paper uses the DNB Household Survey (DHS), a Dutch household panel covering the period 1993–2016.
- The estimation sample is the pre-reform period, 1993–2001.
- The validation sample is the post-reform period, 2006–2016.
- The model focuses on men aged 50 and older because the pension and labour supply decisions are tied to male work histories in the data and because modelling joint household retirement would substantially increase the state space.
- The analysis uses information on labour supply, retirement age, pension claiming, disability insurance, unemployment insurance, health, wages, household wealth, spousal income, and subjective survival expectations.

Interpretation: The key strength of the data is that it combines actual behaviour with subjective probabilities of survival. This allows the model to discipline beliefs directly instead of inferring them from choices alone.

2.2 Institutional setting and pension reform

- During the 1990s, occupational pensions in the Netherlands strongly encouraged early retirement.
- Early occupational pensions could generally be claimed from age 59 without actuarial adjustment.
- Public pensions began automatically at age 65.
- Disability insurance and unemployment insurance provided additional routes out of work before public pension age.
- A 2006 reform effectively prohibited generous early retirement schemes by introducing a tax penalty on early retirement pensions for workers born in 1950 or later.
- After the reform, occupational pensions became actuarially adjusted: claiming earlier lowered benefits and claiming later increased benefits.

Interpretation: The reform creates a demanding external validation exercise. The model is estimated in the old regime and then used to predict behaviour in a substantially different regime.

2.3 Subjective survival expectations

- Respondents report probabilities of surviving to target ages such as 65, 75, 80, 85, 90, 95, and 100.
- Answers are reported on an 11-point scale from 0 to 10, interpreted as probabilities from 0% to 100%.
- The reported probabilities decline monotonically with target age on average.
- There is substantial heterogeneity in reported beliefs, with standard deviations often around 20–30 percentage points.

Interpretation: Subjective survival is not merely an average shift relative to life tables. The dispersion across individuals is central to the paper’s mechanism.

2.4 Three survival belief specifications

1. **Life tables.** Mortality is based on actuarial tables adjusted for current health. This is the standard approach in many empirical life-cycle models.
2. **Average subjective expectations.** The average level of beliefs is adjusted to match reported subjective probabilities, but all individuals share the same survival curve conditional on observables.
3. **Heterogeneous subjective expectations.** Individual-specific survival curves are estimated from reported survival probabilities using a measurement model.

Interpretation: The comparison isolates whether policy forecasts improve because subjective expectations shift average survival or because they add individual-level heterogeneity.

2.5 Life-cycle model outcomes

The model targets moments for:

- annual hours worked;
- labour force participation;
- occupational pension claiming;
- disability insurance claiming;

- unemployment insurance claiming;
- wealth quartiles.

Interpretation: The model is not fitted only to retirement age. It must jointly match labour supply, benefit claiming, and wealth accumulation.

3 Models and empirical specifications

3.1 Retirement choices

Agents aged 50 and older make yearly choices over labour supply, benefit claiming, and consumption. Labour supply takes four values:

$$h_t \in \{0, 1500, 2000, 2500\}. \quad (1)$$

- Labour supply is available before age 70.
- Disability insurance can be claimed only in poor health.
- Unemployment insurance is automatic when the individual does not work, does not claim DI or occupational pension, and is eligible based on work history.
- Occupational pension claiming is available from age 59 to 69 and is absorbing.
- Public pension benefits start automatically at age 65.

Interpretation: The retirement decision is not a single binary retirement choice. Workers choose among work, disability insurance, unemployment insurance, occupational pension claiming, and saving.

3.2 Utility from consumption and leisure

Individuals derive utility from consumption c_t and leisure l_t :

$$u(c_t, l_t) = \frac{n_t}{1 - \sigma} \left[\left(\left(\frac{c_t}{n_t} \right)^\kappa l_t^{1-\kappa} \right)^{1-\sigma} - 1 \right]. \quad (2)$$

- n_t is an equivalence scale based on household size.
- σ governs risk aversion and intertemporal substitution.
- κ determines the relative weight on consumption versus leisure.

Interpretation: Beliefs about longevity affect the value of saving and future leisure. Therefore survival expectations influence the estimated values of σ , κ , and the discount factor.

3.3 Leisure, work costs, and stigma costs

Leisure is defined as:

$$l_t = 4000 - h_t - \psi I\{h_t > 0\} - \delta I\{health = bad\} - \phi I\{claim DI\} - \xi I\{claim UI\}. \quad (3)$$

- ψ is the fixed time cost of working.
- δ is the time cost of bad health.
- ϕ and ξ are stigma costs of disability insurance and unemployment insurance.

Interpretation: These terms rationalize reduced labour supply in poor health and incomplete take-up of social insurance benefits despite eligibility.

3.4 Bequest utility

Bequest utility is:

$$b(w_t, n_t) = \exp[\theta_0 + \theta_1 n_t] \times \frac{(w_t + K)^{(1-\sigma)\kappa}}{1 - \sigma}. \quad (4)$$

- w_t is wealth.
- K controls curvature at low wealth.
- θ_0 and θ_1 govern the strength of bequest motives.
- The dependence on n_t allows bequests to matter more when a spouse is likely present.

Interpretation: Bequest motives are important for matching wealth at older ages in the Dutch institutional setting, where pensions and health insurance reduce some precautionary motives.

3.5 Health transition

Health M_t is binary: good or bad. The transition probability is:

$$Pr(M_{t+1} = 2 \mid M_t, t) = \frac{\exp[\mu_0 + \mu_1 t + \mu_2 I\{M_t = 2\}]}{1 + \exp[\mu_0 + \mu_1 t + \mu_2 I\{M_t = 2\}]}. \quad (5)$$

Interpretation: Health affects labour supply directly through leisure costs and indirectly through disability insurance eligibility.

3.6 Budget constraint

Cash-on-hand before the consumption floor is:

$$x_t = w_t + \tau(earn_t, DIinc_t, UIinc_t, pubpens_t, occpens_t, w_t, t) + inc_t^{sp} - OOP_t. \quad (6)$$

The consumption floor is:

$$\tilde{x}_t = \max\{x_t, n_t c_{min}\}. \quad (7)$$

- $\tau(\cdot)$ maps earnings, transfers, pensions, and wealth into net resources.
- inc_t^{sp} is spousal income.
- OOP_t is out-of-pocket medical expenditure.
- c_{min} is the minimum consumption level associated with social assistance.

Interpretation: The budget constraint embeds the Dutch tax and transfer system, so the same preference estimates can be used to simulate behaviour under the post-reform pension rules.

3.7 Measurement model for subjective survival

Reported survival probabilities are noisy and rounded. Let S_{isk} be the latent Gompertz survival probability for individual i , survey wave s , and target age k . The latent reported probability is:

$$P_{isk}^* = S_{isk} + \varepsilon_{isk}. \quad (8)$$

The observed report P_{isk} is a rounded version of P_{isk}^* :

$$P_{isk} = Round_m(P_{isk}^*), \quad m \in \{10, 50, 100\}. \quad (9)$$

- A report of 60 can only come from rounding to multiples of 10.
- Reports of 0 or 100 can arise from several rounding levels.
- Recall error is heteroskedastic and depends on observed characteristics.
- Individual-level unobserved heterogeneity generates variation in subjective survival curves.

Interpretation: The measurement model prevents coarse survey reports from being treated as exact probabilities. It extracts individual-specific survival beliefs from noisy, rounded answers.

3.8 Estimation and validation

Preferences are estimated by simulated method of moments in the pre-reform sample. The model then simulates behaviour under post-reform pension rules.

$$\hat{\theta} = \arg \min_{\theta} [m^{data} - m^{sim}(\theta)]' W [m^{data} - m^{sim}(\theta)]. \quad (10)$$

Interpretation: The key test is not whether the model fits the old regime. All three models do. The test is whether the estimated model correctly predicts behaviour after the pension reform.

4 Identification and empirical design

4.1 Why expectations matter for identification

- Preferences and beliefs both affect savings and retirement decisions.
- Without direct belief data, a structural model can explain the same behaviour through different combinations of preferences and survival expectations.
- Subjective survival reports discipline beliefs directly.
- This changes preference estimates even when within-sample fit remains similar.

Interpretation: The paper’s central point is that within-sample fit is not a sufficient validation criterion for structural models when expectations are misspecified.

4.2 External validation through pension reform

- The model is estimated using 1993–2001 data.
- A large policy reform changed occupational pensions in the 2000s.
- The model is used to predict retirement behaviour in 2006–2016.
- Observed retirement ages in the post-reform period provide an out-of-sample benchmark.

Interpretation: The reform creates a hard test of the model because it changes incentives in a way that the estimator did not use when choosing preferences.

4.3 What makes the heterogeneous-expectations model different

- The heterogeneous model adds only one state variable that summarizes individual-specific subjective survival.
- It does not rely on flexible latent preference types.

- It links variation in behaviour to observed survey reports about survival.
- It therefore improves policy predictions without simply adding unrestricted heterogeneity.

Interpretation: The improvement is economically disciplined: the model uses measured beliefs rather than black-box preference heterogeneity.

4.4 Limitations

- Survival expectations are assumed exogenous and stable.
- The model does not examine whether subjective expectations are rational.
- The model focuses on male retirement and abstracts from joint household labour supply decisions.
- The model does not include employer switching or detailed portfolio choice.
- The post-reform validation period is not a randomized experiment, although the reform provides strong institutional variation.

Interpretation: The paper's strongest claim is not that all expectations are correct, but that people appear to behave as if their subjective expectations matter.

5 Tables and figures

Visuals are directly extracted from the paper and grouped compactly. Each item follows the same *Read* plus *Economic message* structure.

5.1 Figure 1 and Table 1: decisions and survival data

Read:

- Figure 1 shows the choice set at different ages: labour supply, disability insurance, unemployment insurance, occupational pension claiming, public pension receipt, and consumption/saving.
- Discrete choices are concentrated before age 70; after that, individuals no longer supply labour in the model.
- Table 1 shows reported survival probabilities in the pre- and post-reform samples.
- Reported probabilities decline with target age but exhibit substantial dispersion.
- The post-reform sample has more observations and similar average subjective probabilities at most target ages.

Economic message: The model studies retirement as a sequence of linked choices. Subjective survival expectations provide the belief input that disciplines those choices.

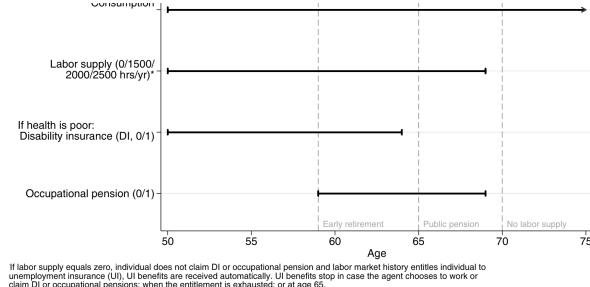


FIGURE 1
Decisions available at various ages

Figure 1: decisions available at various ages

TABLE 1
Descriptive statistics of reported survival probabilities (0–100%)

	Current age	Pre-reform sample (1993–2001)			Post-reform sample (2006–16)		
		N	Mean	Std. Dev.	N	Mean	Std. Dev.
Target age 65	50–65	–	–	–	879	79	16
Target age 75	50–65	1,629	69	20	3,706	70	19
Target age 80	50–70	2,008	56	23	5,169	59	23
Target age 85	65–75	646	48	22	2,505	55	23
Target age 90	70–80	381	35	24	1,792	41	25
Target age 95	75–85	151	27	23	1,185	30	24
Target age 100	80–89	43	26	30	553	18	21

Table 1: reported survival probabilities

5.2 Figure 2: measurement model for subjective survival

Read:

- The black curve is the individual-specific Gompertz survival curve.
- Empty circles are reported survival probabilities.
- Vertical normal distributions capture recall error around latent survival probabilities.
- Shaded regions show how different levels of rounding map latent probabilities into observed responses.

Economic message: Survey reports are coarse and noisy, but the measurement model extracts a full individual survival curve from them.

5.3 Table 2: preference estimates

Read:

- The life-table and average-subjective-expectations models estimate similar risk aversion, consumption weight, and discount factor.

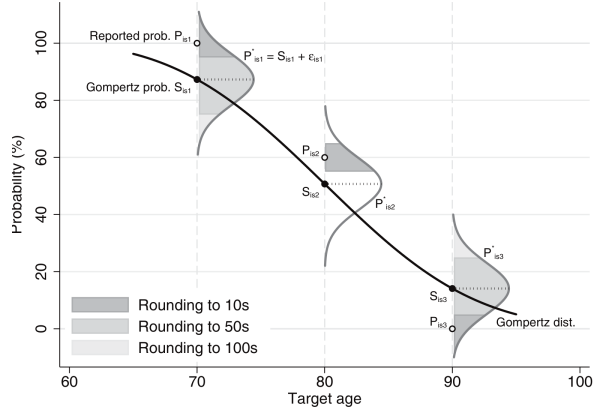


FIGURE 2
Illustration of the logic of the rounding model

vels (a “5” may be rounded to 10 s or 50 s and a “0” or “10” can result from all three levels).

TABLE 2
Estimates for preference parameters

	(1) Life tables	(2) Average subj. exp.	(3) Heterogeneous exp.
Utility function ^a			
σ (concavity)	4.64 (0.000023)	4.64 (0.000028)	5.18 (0.000038)
κ (consumption weight)	0.68 (0.000061)	0.65 (0.000063)	0.42 (0.000065)
β (discount factor)	1.04 (0.000026)	1.04 (0.000035)	0.98 (0.000028)
ψ (fixed cost of work; hours/year)	1085 (27.5)	1013 (4.5)	1101 (8.0)
δ (time cost of bad health; hours/year)	131 (13.9)	249 (34.6)	251 (6.9)
ϕ (stigma cost DI; hours/year)	1681 (441.9)	2029 (345.5)	2140 (145.4)
ζ (stigma cost UI; hours/year)	3620 (8.9)	3509 (1.9)	3568 (5.9)
Bequests ^b			
θ_0 (intercept bequest weight)	-2.45 (0.00014)	-3.91 (0.00015)	-9.27 (0.00010)
θ_1 (slope bequest weight)	4.73 (0.00012)	5.56 (0.00014)	2.51 (0.000082)
K (concavity bequests)	675.282 (29.0)	913.915 (34.3)	441.061 (6.0)
Function value	473.27	473.21	483.89

Notes: Standard errors in parentheses. ^a $u(c_t, l_t) = \frac{\theta_0}{1-\sigma} [((\frac{c_t}{l_t})^\kappa l_t^{1-\kappa})^{1-\sigma} - 1]$; $l_t = 4,000 - h_t - \psi \mathbb{I}(h_t > 0) - \delta \mathbb{I}(\text{health} = \text{bad}) - \phi \mathbb{I}(\text{claim DI}) - \zeta \mathbb{I}(\text{claim UI})$. ^b $b(w_t, n_t) = \exp(\theta_0 + \theta_1 n_t) \times \frac{(w_t + K)^{(1-\sigma)\kappa}}{1-\sigma}$.

- The heterogeneous-expectations model estimates higher concavity $\sigma = 5.18$.
- It estimates a lower consumption weight $\kappa = 0.42$, meaning leisure receives relatively more weight.
- It estimates a lower and more plausible discount factor $\beta = 0.98$ rather than 1.04.
- Bequest parameters also differ substantially across specifications.

Economic message: Similar behaviour in the pre-reform regime can be rationalized by different combinations of beliefs and preferences. Heterogeneous beliefs change the estimated preference structure.

5.4 Figure 3: within-sample model fit

Read:

- All three models fit pre-reform labour supply and participation reasonably well.
- Benefit claiming patterns are matched closely, especially occupational pension claiming around ages 59–64.
- Wealth quartiles are matched well through ages 70–74, after which data uncertainty becomes larger.
- The model with heterogeneous expectations does not obviously dominate within sample.

Economic message: Within-sample fit alone cannot reveal which expectations model is correct. Counterfactual validation is needed.

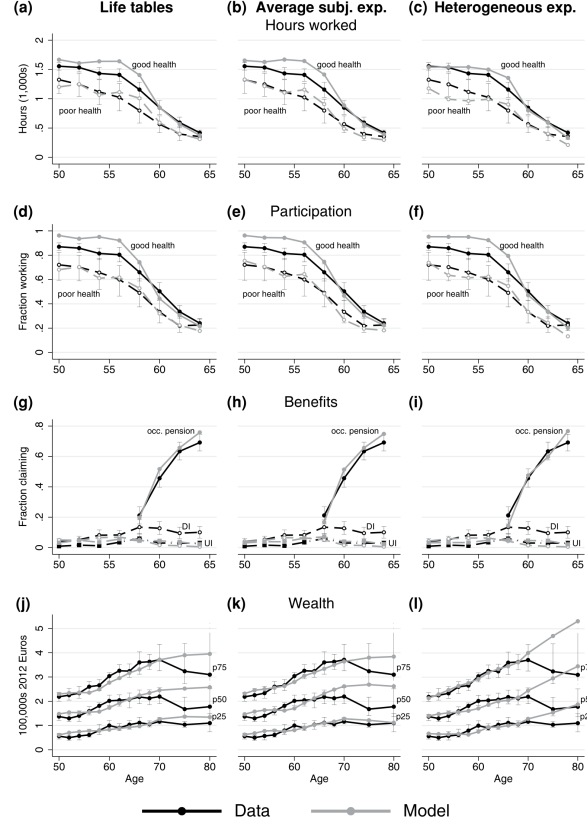


FIGURE 3
Labor market and retirement profiles

5.5 Table 3 and Figure 4: counterfactual pension reform predictions

Read:

- Table 3 shows that the observed average retirement age rises from 61.2 to 63.8, a difference of 2.6 years.
- The life-table model predicts an increase of 5.2 years, from 60.7 to 65.9.
- The average-subjective-expectations model predicts an increase of 4.2 years, from 60.9 to 65.1.
- The heterogeneous-expectations model predicts an increase of 2.7 years, from 61.0 to 63.7.
- Figure 4 shows the same result in labour supply profiles: homogeneous-expectations models overstate post-reform labour supply at ages 60–63.

Economic message: Heterogeneous subjective expectations are crucial for external validity. They turn an overprediction of the policy response into a near match with the observed retirement shift.

TABLE 3
Average retirement age with ('93-'01) and without ('06-'16) generous early retirement schemes

	Pre-reform (1993–2001)	Post-reform (2006–16)	Difference (years)
a. Data			
Average pension age ^{a,b}	61.2 (60.97–61.41)	63.8 (63.49–64.03)	2.6 (2.23–2.92)
N	756	467	1223
b. Simulations			
Life tables	60.7	65.9	5.2
Average subj. exp.	60.9	65.1	4.2
Heterogeneous exp.	61.0	63.7	2.7

Notes: ^aAge of retirement combines information from three sources: (1) actual retirement age reported by retirees; (2) observed retirement age if respondent retires while in the sample; (3) expected retirement age if neither reported nor observed retirement age is available. ^bRetirement ages below 50 or above 70 are set to missing; less than 1% of observations are dropped this way. 95% confidence intervals based on robust standard errors in parentheses.

Table 3: average retirement age before and after reform

5.6 Figure 5: behaviour conditional on survival beliefs before the reform

Read:

- Figure 5 separates individuals who expect long lives from those who expect short lives.
- In the pre-reform data, differences across subjective survival groups are modest for hours, participation, and pension claiming.
- The model with heterogeneous expectations captures many of these pre-reform differences.
- Wealth differences by survival expectations are also informative because bequest motives and survival beliefs jointly shape saving.

Economic message: Before the pension reform, generous early retirement muted the behavioural consequences of survival heterogeneity. Beliefs matter more after incentives change.

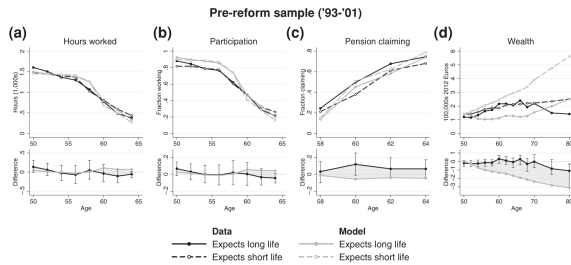


FIGURE 5

Behaviour conditional on survival expectations for the pre-reform sample ("long life" means top 45% of distribution of subjective survival, "short life" means bottom 45%)

reform. Changing only the institutional rules for occupational pensions, the model with heterogeneous expectations continues to predict an average retirement age of 63.7, about 1 month younger than the average age in the data.

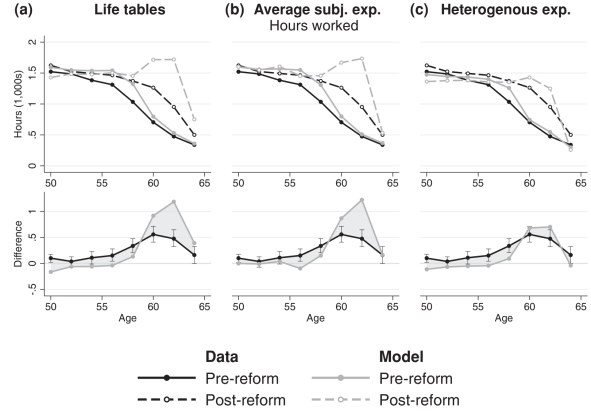


FIGURE 4

Simulated and observed labour supply in the pre-reform (1993–2001) and post-reform (2006–16) samples

The differences in labour supply between data and simulations at ages 60–63 are less than half as large as those in panels (a) and (b). The bottom graph illustrates that the before-after com-

Figure 4: observed and simulated labour supply

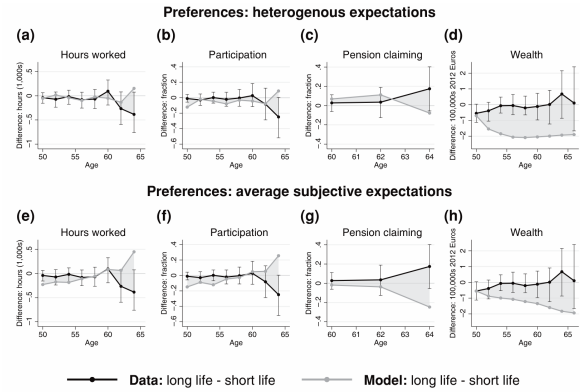


FIGURE 6

Differences in behaviour across the distribution of survival expectations for the post-reform sample ("long life" means top 45% of distribution of subjective survival, "short life" means bottom 45%)

5.7 Figure 6: behaviour conditional on survival beliefs after the reform

Read:

- Figure 6 compares long-life and short-life groups in the post-reform sample.

- With preferences estimated under heterogeneous expectations, simulated differences are closer to the data.
- With preferences from the average-subjective-expectations model, differences across survival groups become too large or move in the wrong direction.
- The post-reform regime makes subjective survival more behaviorally relevant because pensions are actuarially adjusted.

Economic message: The heterogeneous-expectations model improves not only the average retirement prediction but also behaviour conditional on beliefs.

5.8 Figure 7: why preference estimates matter for predictions

Read:

- Figure 7 varies preference parameters one at a time while holding heterogeneous beliefs fixed.
- Risk aversion, consumption weight, the discount factor, and bequest parameters all have large effects on the simulated post-reform retirement age.
- Parameters from homogeneous-belief models imply an excessively strong retirement response to actuarial adjustments.
- The heterogeneous-expectations preference estimates produce a simulated retirement age close to the data.

Economic message: Belief misspecification affects policy predictions indirectly through preference estimates. The wrong expectations model gives the wrong preferences for counterfactuals.

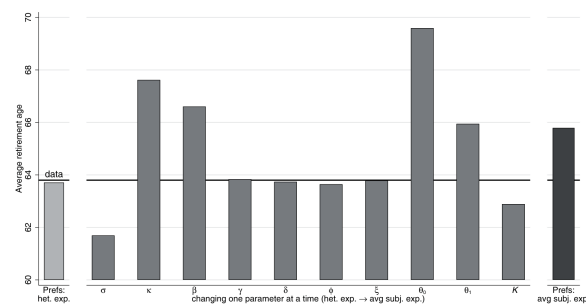


FIGURE 7
The impact of changing preference parameters on counterfactual simulations for the model with heterogeneous expectations

6 Short summary

- The paper asks whether individual-level subjective survival expectations improve structural counterfactual predictions.
- It estimates a dynamic life-cycle model of Dutch retirement, benefit claiming, and saving.
- The model is estimated using pre-reform data from 1993–2001.
- It is validated using post-reform data from 2006–2016 after actuarial adjustments to occupational pensions.
- Three survival specifications are compared: life tables, average subjective expectations, and heterogeneous subjective expectations.
- All three models fit pre-reform data similarly well.
- Preference estimates differ sharply across beliefs specifications.

- Homogeneous-expectations models overpredict the post-reform increase in retirement age.
- The heterogeneous-expectations model predicts an increase of 2.7 years, close to the observed 2.6-year increase.
- The key lesson is that expectations can matter for counterfactuals even when they do not visibly change within-sample fit.

7 Conclusion

7.1 Core conclusion

Individual-specific survival expectations derived from subjective survey data improve the out-of-sample predictions of a structural life-cycle retirement model. The improvement is not primarily visible in within-sample fit. It appears when the model is asked to predict behaviour under a new policy regime.

7.2 Economic intuition

Workers' retirement and saving decisions depend on how long they expect to live. If a model forces everyone to share the same survival beliefs, it must use preferences to rationalize behaviour that is partly driven by heterogeneous beliefs. Those distorted preference estimates can still fit the old regime but perform poorly in a new regime. Heterogeneous subjective beliefs correct the preference estimates and improve policy forecasts.

7.3 Takeaway

The paper provides a clear lesson for structural modelling: external validity requires getting expectations right. Subjective expectations are not just descriptive survey variables; they can materially improve counterfactual policy evaluation.